Fall 2026 DSC 242: Homework 5

Due 11:59PM Friday, November 6

Submission. Parts 2(c) and 3(b) are experimental: submit them as a single PDF report (for example, exported from a Jupyter notebook) that contains your figures, a short discussion for each part, and the code that produced the figures (as notebook cells or as an appendix). Every figure should have labeled axes, a legend, and a caption stating the parameters and the number of trials. The other parts are written problems.

1. (Vershynin 5.21, matrix Hoeffding inequality, 30 points) Let $\epsilon_1, \dots, \epsilon_n$ be i.i.d. Rademacher random variables, and $A_1, \dots, A_n \in \mathbb{R}^{d \times d}$ be any deterministic symmetric matrices. Show that for any $t \geq 0$,

$$\mathbb{P}\Big[\Big\|\sum_{i=1}^n \epsilon_i A_i\Big\|_{\text{op}} \geq t\Big] \leq 2d \exp\Big(-\frac{t^2}{2v}\Big)$$

   where $v = \|\sum_{i=1}^n A_i^2\|_{\text{op}}$.

   [Hint: Apply Lieb's inequality together with a method to bound $\mathbb{E}[e^{\lambda \epsilon_i A_i}]$.]

2. (Stochastic block model, 35 points) Suppose n is even, and let $0 < q < p < 1$. Let $A \in \{0,1\}^{n \times n}$ be a symmetric random matrix such that $\{A_{ij}\}_{1 \leq i \leq j \leq n}$ are independent with

$$A_{ij} \sim \begin{cases} \text{Bernoulli}(p) & \text{if } i \leq j \leq n/2 \text{ or } n/2+1 \leq i \leq j \\ \text{Bernoulli}(q) & \text{if } i \leq n/2 \text{ and } j \geq n/2+1 \end{cases}$$

   (A is the adjacency matrix of a random graph with two communities of $n/2$ vertices, where vertices within the same community are more likely to be connected by an edge.)

   (a) (10 points) Show that $\mathbb{E}[A] \in \mathbb{R}^{n \times n}$ is a rank-2 matrix, and compute its two non-zero eigenvalues in terms of n, p, q.

   (b) (15 points) Apply the matrix Bernstein inequality to show that for a universal constant $c > 0$ and any $t \geq 0$,

$$\mathbb{P}[\|A - \mathbb{E}[A]\|_{\text{op}} \geq t] \leq 2n \exp\Big(-c \min\Big(\frac{t^2}{d}, t\Big)\Big)$$

   where $d = n \cdot \frac{p+q}{2}$ is the expected average vertex degree. Supposing that $d \gg \log n$, for what (n-dependent) pairs of values $p > q$ do you expect $\|A - \mathbb{E}[A]\|_{\text{op}}$ to be smaller than the 2-nd eigenvalue of $\mathbb{E}[A]$ with high probability for large n?

   (c) (Experiment: spectral clustering, 10 points) Generate A exactly as above (including the diagonal) with $n = 1000$ vertices and expected average degree $d = \frac{n(p+q)}{2} = 50$ fixed, and vary the gap $\Delta = \frac{n(p-q)}{2} = \lambda_2(\mathbb{E}A)$ over $\Delta/\sqrt{d} \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 3\}$, i.e., $p = \frac{d+\Delta}{n}$ and $q = \frac{d-\Delta}{n}$. For each Δ record, over at least 20 trials, $\|A - \mathbb{E}A\|_{\text{op}}$, the

eigenvalues $\lambda_2(A) \geq \lambda_3(A)$ (eigenvalues in decreasing order), and the misclassification rate of *spectral clustering*: assign vertex i to a community according to the sign of the i-th coordinate of the eigenvector of $\lambda_2(A)$, and report the fraction of wrongly assigned vertices, minimized over the two ways of naming the communities (so it lies in $[0, \frac{1}{2}]$). Plot in the left panel the medians of $\|A - \mathbb{E}A\|_{\mathrm{op}}$, $\lambda_2(A)$ and $\lambda_3(A)$ against $\Delta/\sqrt{d}$, with error bars showing the quartiles, together with the constant $2\sqrt{d}$, the line Δ, and the *Bernstein level*

$$t_{\mathrm{B}}(v, b, d) = \sqrt{2v \log(4d)} + \frac{2b}{3} \log(4d),$$

at which the matrix Bernstein inequality for a sum S of $d \times d$ matrices with parameters v, b gives $\mathbb{P}[\|S\|_{\mathrm{op}} \geq t_{\mathrm{B}}] \leq \frac{1}{2}$ (check this), so that the median of $\|S\|_{\mathrm{op}}$ is at most t_{B}; here use $t_{\mathrm{B}}(v, 1, n)$ with $v = \frac{n}{2}\big(p(1-p) + q(1-q)\big)$ and $b = 1$, the exact parameters of the summands of $A - \mathbb{E}A$ in part (b). In the right panel plot the average misclassification rate against $\Delta/\sqrt{d}$ (format: Figure 1; 4 points). Then discuss:

i. (2 points) Is $\|A - \mathbb{E}A\|_{\mathrm{op}}$ close to $2\sqrt{d}$, or to the bound t_{B} of part (b)?

ii. (4 points) Part (b) asks for $\Delta > \|A - \mathbb{E}A\|_{\mathrm{op}}$, and by Weyl's inequality $|\lambda_k(A) - \lambda_k(\mathbb{E}A)| \leq \|A - \mathbb{E}A\|_{\mathrm{op}}$ this implies $\lambda_2(A) \geq \Delta - \|A - \mathbb{E}A\|_{\mathrm{op}}$ and $\lambda_3(A) \leq \|A - \mathbb{E}A\|_{\mathrm{op}}$. In your plot, at what value of $\Delta/\sqrt{d}$ does $\lambda_2(A)$ separate from $\lambda_3(A)$, and at what value does the misclassification rate fall below 10%? Compare these two thresholds with the three conditions $\Delta > t_{\mathrm{B}}$ (part (b) with the Bernstein bound), $\Delta > \|A - \mathbb{E}A\|_{\mathrm{op}}$ (part (b) with the true norm) and $\Delta > \sqrt{d}$. Which of them is closest to what you observe, and is the condition of part (b) necessary for spectral clustering to work?

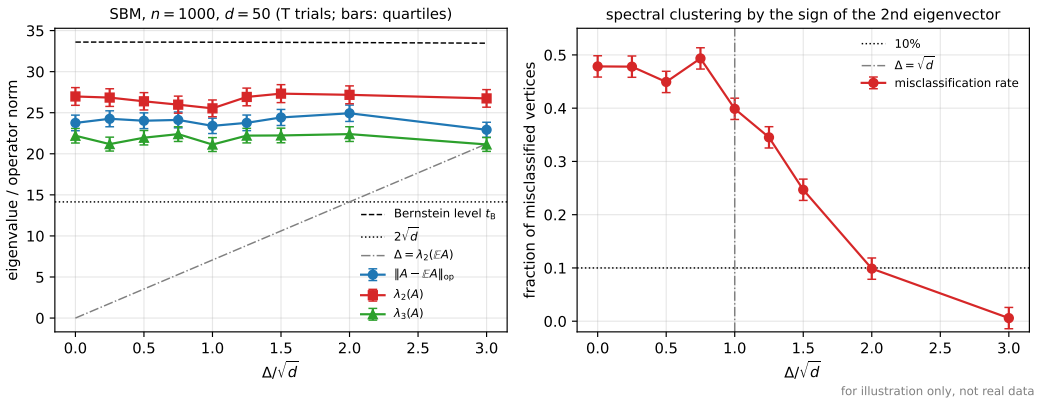


Figure 1: Expected format of the figure for part 2(c), for illustration only (not real data).

3. (The logarithm in covariance estimation, 35 points) Let $X_1, \ldots, X_n \in \mathbb{R}^d$ be i.i.d. random vectors and

$$\widehat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} X_i X_i^\top$$

be the sample covariance matrix.

(a) (25 points) Suppose that each $X_i \sim \mathrm{Uniform}(\pm\sqrt{d}e_1, \ldots, \pm\sqrt{d}e_d)$, where $e_1, \ldots, e_d$ are the standard basis vectors in $\mathbb{R}^d$. Show that when $n = d$, there is a sufficiently small

constant $c > 0$ for which

$$\mathbb{P}\left[\|\widehat{\Sigma}\|_{\mathrm{op}} \geq \frac{c \log n}{\log \log n}\right] \to 1 \text{ as } n \to \infty.$$

You may use without proof the following lower bound on the binomial tail, which matches the Chernoff bound $\mathbb{P}[\mathrm{Bin}(n,p) \geq nq] \leq e^{-nD(q\|p)}$ up to the polynomial prefactor, where $D(q\|p) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}$: for any $q > p$ such that nq is an integer,

$$\mathbb{P}[\mathrm{Bin}(n,p) \geq nq] \geq \frac{1}{\sqrt{8nq(1-q)}} e^{-nD(q\|p)}.$$

Thus the logarithmic factors in the matrix Bernstein bound for $\|\widehat{\Sigma} - \Sigma\|_{\mathrm{op}}$ cannot be removed in general.

(b) (Experiment: sparse versus spherical, 10 points) Consider two distributions for the X_i:

(A) *Sparse*: $X_i \sim \mathrm{Uniform}(\pm\sqrt{d}e_1, \ldots, \pm\sqrt{d}e_d)$, as in part (a). Here $\widehat{\Sigma}$ is diagonal with entries $\frac{d}{n}N_j$, where $(N_1, \ldots, N_d) \sim \mathrm{Multinomial}(n; \frac{1}{d}, \ldots, \frac{1}{d})$ are the numbers of samples falling on each axis, so $\|\widehat{\Sigma} - I\|_{\mathrm{op}} = \max_j |\frac{d}{n}N_j - 1|$ can be computed from the counts without forming any matrix.

(B) *Spherical*: $X_i \sim \mathrm{Uniform}(\sqrt{d}\, S^{d-1})$, i.e., $X_i = \sqrt{d}\, g_i/\|g_i\|_2$ with $g_i \sim \mathcal{N}(0, I_d)$.

Both have $\mathbb{E}X_iX_i^\top = I$ and $\|X_i\|_2^2 = d$, so $\widehat{\Sigma} - I = \sum_{i=1}^n Y_i$ with $Y_i = \frac{1}{n}(X_iX_i^\top - I)$ has for both the *same* parameters

$$b = \|Y_i\|_{\mathrm{op}} = \frac{d-1}{n}, \qquad v = \Big\|\sum_{i=1}^n \mathbb{E}Y_i^2\Big\|_{\mathrm{op}} = \frac{d-1}{n},$$

since $X_iX_i^\top - I$ has eigenvalues $d-1$ (once) and -1 ($d-1$ times), and $\mathbb{E}[(X_iX_i^\top - I)^2] = \mathbb{E}[\|X_i\|_2^2 X_iX_i^\top] - I = (d-1)I$. Consider two regimes: (i) $n = d$ with $d \in \{16, 32, \ldots, 4096\}$, where $b \log d$ is the dominant term of the bound; (ii) $n = 50d$ with $d \in \{16, 32, \ldots, 1024\}$, where $\sqrt{v \log d}$ dominates. Plot the median of $\|\widehat{\Sigma} - I\|_{\mathrm{op}}$ over at least 20 trials against d on log-log axes (error bars: quartiles), one panel per regime, one curve per distribution, together with the Bernstein level $t_{\mathrm{B}}(v, b, d)$ of part 2(c) and the reference curves $\frac{\log d}{\log \log d}$ in regime (i) and $2\sqrt{d/n}$ and $\sqrt{2(d/n)\log d}$ in regime (ii) (format: Figure 2; 4 points). Then discuss:

i. (3 points) The two distributions have the same v and b, hence the same Bernstein level. Which one follows the growth of the bound and which one stays bounded (regime (i)) or follows $2\sqrt{d/n}$ (regime (ii))? For (A), explain the growth in regime (i) through part (a) (the maximum load of n balls thrown into $d = n$ bins) and in regime (ii) through a maximum of d approximately Gaussian variables (what is the typical size of the maximum of d independent $\mathcal{N}(0, \sigma^2)$ variables?).

ii. (3 points) The two distributions have the same v, b and d, so the matrix Bernstein inequality, and more generally any bound of the form "$\|S\|_{\mathrm{op}} \leq C(v, b, d)$ with probability at least $\frac{1}{2}$" that depends on the summands only through v, b and d, gives the same number for both. Using what you observed for (A) in regime (ii), how large must $C(v, b, d)$ be as a function of v and d? For the spherical distribution (B) in regime (ii), the truth is $\|\widehat{\Sigma} - I\|_{\mathrm{op}} \approx 2\sqrt{d/n} = 2\sqrt{v}$. By how much is any bound of this form off for (B)? Conclude that a proof of $\|\widehat{\Sigma} - I\|_{\mathrm{op}} \lesssim \sqrt{v}$ for (B) must use some property of the X_i other than v and b; which property distinguishes (B) from (A)?

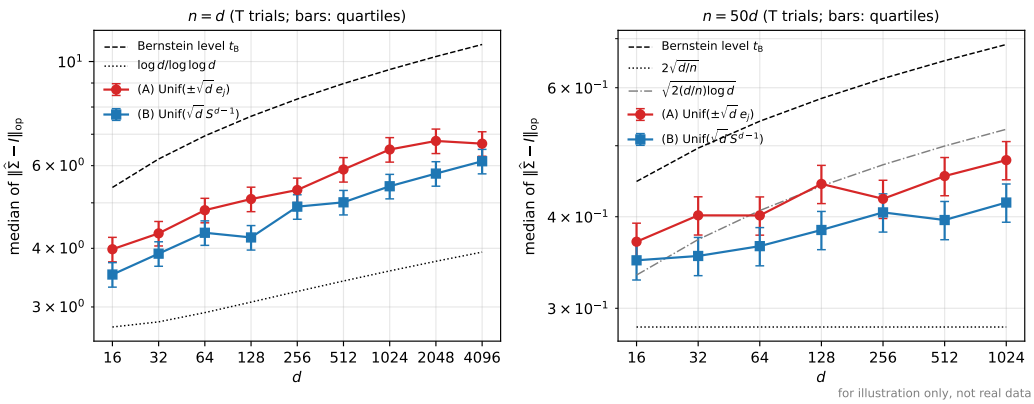


Figure 2: Expected format of the figure for part 3(b), for illustration only (not real data).